Routing Scheme Evaluation Results

Grid Networks with csma_bianchi Interference

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1 Evaluation Setup

Parameter	Value
Seeds per combo	30 (seeds 1–30); values are mean ± std makespan
Schedulers	HEFT: calibrated to runtime routing model
	HEFT-1: direct-link BW; 0.001 MB/s for non-adjacent
	HEFT-2: widest-path BW for all pairs (fixed)
Routing schemes	W, S, SH, GS, GC, GB, GO, GSD, GSD-D
Interference model	`csma_bianchi` (802.11ax, 5 GHz, 20 MHz)
RF parameters	$P_{\mathrm{tx}} = 20\,\mathrm{dBm}$, path-loss exponent $n = 3.0$
Grid spacing	40 m; adjacent-link PHY rate ≈ 8.6 MB/s
Compute costs	150–1000 cu (heterogeneous, ≈ 6.7× range)
Data sizes	2–30 MB (heterogeneous, 15× range)
Node capacities	80–300 cu/s (heterogeneous)
Total runs	3 schedulers × 9 routing × 4 experiments × 30 seeds = 3240

Table 1: Evaluation parameters.

HEFT (calibrated) passes the runtime routing model to HEFT's pairwise rate matrix, so placement decisions are calibrated to whichever routing scheme runs at simulation time. Equals HEFT-2 when paired with `widest_path`.

HEFT-1 pins 0.001 MB/s on all non-adjacent pairs regardless of runtime routing. Strongly biases placement toward same-node or direct-neighbour assignments.

HEFT-2 always uses `WidestPathRouting` for pairwise BW estimates, regardless of the runtime routing scheme.

SH (Shortest Hop) minimises hop count, breaking ties by $\sum 1/b_\ell$. Added alongside S to test whether reducing relay count reduces csma_bianchi interference enough to offset lower bottleneck bandwidth.

Extra metrics: H = mean path length (hops) per transfer; PLU = peak link utilisation (fraction of time the busiest link was active). Values are averaged over 30 seeds.

2 Full Results Tables

Mean makespan ± std in seconds, over 30 seeds. **Bold green**: global experiment winner. **Bold**†: best within that scheduler column. Red: worst within that scheduler column. H = mean hops per transfer; PLU = peak link utilisation.

2.1 4×4 Grid, Small DAG (8 tasks, fork-join, 12 edges)

Routing	HEFT (calib.)			HEFT-1			HEFT-2		
	ms (s)	H	PLU	ms (s)	H	PLU	ms (s)	H	PLU
W	1814.518 ± 0.000	6.86	0.01	18.336 ± 0.000	1.00	0.13	1814.518 ± 0.000	6.86	0.01
S	93.856 ± 0.000	2.00	0.09	18.336 ± 0.000	1.00	0.13	93.856 ± 0.000	2.00	0.09
SH	93.856 ± 0.000	2.00	0.09	18.336 ± 0.000	1.00	0.13	93.856 ± 0.000	2.00	0.09
GS	93.319 ± 0.000	2.00	0.14	18.336 ± 0.000	1.00	0.13	93.319 ± 0.000	2.00	0.14
GC	93.319 ± 0.000	2.00	0.14	18.336 ± 0.000	1.00	0.13	93.319 ± 0.000	2.00	0.14
GB	93.319 ± 0.000	2.00	0.14	18.336 ± 0.000	1.00	0.13	93.319 ± 0.000	2.00	0.14
GO	93.319 ± 0.000	2.00	0.14	18.336 ± 0.000	1.00	0.13	93.319 ± 0.000	2.00	0.14
GSD	140.483 ± 0.000	6.86	0.07	18.336 ± 0.000	1.00	0.13	140.483 ± 0.000	6.86	0.07
GSD-D	**91.134**† ± 0.000	6.86	0.15	**18.035** ± 0.000	1.00	0.13	**91.134**† ± 0.000	6.86	0.15
Best	*GSD-D 91.134 ±*			*GSD-D 18.035 ±*			*GSD-D 91.134 ±*		

Table 2: 4×4 small. HEFT-1 co-locates all 8 tasks on one node; no transfers occur so routing is irrelevant (all schemes tie near 18 s). GSD-D gains a small margin via deferral. HEFT and HEFT-2 are identical: the $4{\times}4$ topology's BW estimates converge across routing models. Std devs are near zero because same-node execution is deterministic.

2.2 4×4 Grid, Large DAG (30 tasks, 5-stage pipeline, 48 edges)

Routing	HEFT (calib.)			HEFT-1			HEFT-2		
	ms (s)	H	PLU	ms (s)	H	PLU	ms (s)	H	PLU
W	2383.795 ± 0.000	4.57	0.01	497.788 ± 0.000	1.17	0.03	2383.795 ± 0.000	4.57	0.01
S	1071.723 ± 0.000	1.90	0.01	292.425 ± 0.000	1.03	0.05	1071.723 ± 0.000	1.90	0.01
SH	1071.723 ± 0.000	1.90	0.01	292.425 ± 0.000	1.03	0.05	1071.723 ± 0.000	1.90	0.01
GS	1026.909 ± 0.000	2.21	0.01	**292.091** ± 0.000	1.03	0.04	1026.909 ± 0.000	2.21	0.01
GC	1066.630 ± 0.000	2.29	0.01	292.091 ± 0.000	1.03	0.04	1066.630 ± 0.000	2.29	0.01
GB	1156.735 ± 0.000	2.43	0.01	292.091 ± 0.000	1.03	0.04	1156.735 ± 0.000	2.43	0.01
GO	1022.295 ± 0.000	2.68	0.01	292.091 ± 0.000	1.03	0.04	1022.295 ± 0.000	2.68	0.01
GSD	**855.880**† ± 0.000	4.57	0.01	671.422 ± 0.000	1.17	0.02	**855.880**† ± 0.000	4.57	0.01
GSD-D	1037.424 ± 0.000	4.57	0.01	322.536 ± 0.000	1.17	0.04	1037.424 ± 0.000	4.57	0.01
Best	*GSD 855.880 ±*			*GS 292.091 ±*			*GSD 855.880 ±*		

Table 3: 4×4 large. Under HEFT-1, static greedy schemes (GS–GO) tie at ≈ 292 s (-41% vs W) by routing direct-neighbour transfers efficiently; small std devs confirm stability. Within original HEFT, GSD wins at ≈ 856 s: calibrated placement spreads tasks and GSD's runtime congestion avoidance pays off. Mean hops > 1 under HEFT/HEFT-2 reflect multi-hop paths across the $4{\times}4$ grid.

2.3 7×7 Grid, Small DAG (8 tasks, fork-join, 12 edges)

Routing	HEFT (calib.)			HEFT-1			HEFT-2		
	ms (s)	H	PLU	ms (s)	H	PLU	ms (s)	H	PLU
W	1926.814 ± 0.000	7.22	0.01	24.172 ± 1.405	1.50	0.11	1926.814 ± 0.000	7.41	0.01
S	$\mathbf{93.654}^{\dagger} \pm 0.469$	2.00	0.09	20.262 ± 0.000	1.00	0.17	93.902 ± 0.606	2.00	0.09
SH	93.984 ± 0.624	2.00	0.09	20.262 ± 0.000	1.00	0.17	$\mathbf{93.819}^{\dagger} \pm 0.577$	2.00	0.09
GS	1301.270 ± 494.850	6.86	0.02	20.262 ± 0.000	1.00	0.17	1408.704 ± 422.831	7.35	0.01
GC	1151.118 ± 520.253	6.01	0.02	20.262 ± 0.000	1.00	0.17	1305.283 ± 521.691	6.66	0.02
GB	1503.555 ± 299.095	7.41	0.01	20.262 ± 0.000	1.00	0.17	1495.158 ± 260.578	7.35	0.01
GO	1506.802 ± 269.750	7.46	0.01	20.262 ± 0.000	1.00	0.17	1441.443 ± 260.772	6.81	0.01
GSD	1575.082 ± 253.427	8.17	0.01	20.262 ± 0.000	1.50	0.17	1327.283 ± 403.253	6.99	0.01
GSD-D	1381.950 ± 481.438	7.49	0.01	$\mathbf{19.779} \pm 0.000$	1.50	0.18	1280.647 ± 540.224	7.03	0.01
Best	*S 93.654 ±*			*GSD-D 19.779 ±*			*SH 93.819 ±*		

Table 4: 7×7 small. With 8 tasks and heterogeneous compute, the workload remains largely compute-bound: SH and S both achieve ≈ 94 s under HEFT/HEFT-2. Greedy static schemes (GS–GO) are much worse under HEFT/HEFT-2 because calibrated placement spreads tasks, then static interference-aware routing cannot handle the resulting congestion. GSD-D wins under HEFT-1 (deferral advantage on same-node handoffs). Peak link utilisation is low throughout, confirming compute dominance.

2.4 7×7 Grid, Large DAG (60 tasks, 6-stage pipeline, 102 edges)

Routing	HEFT (calib.)			HEFT-1			HEFT-2		
	ms (s)	H	PLU	ms (s)	H	PLU	ms (s)	H	PLU
W	5638.899 ± 450.976	6.40	0.01	1346.329 ± 256.499	1.47	0.02	5797.983 ± 478.072	6.44	0.01
S	$\mathbf{2461.908}^{\dagger} \pm 271.335$	3.26	0.01	618.534 ± 220.416	1.09	0.05	$\mathbf{2542.188}^{\dagger} \pm 256.661$	3.34	0.01
SH	2537.143 ± 233.861	3.29	0.01	665.738 ± 203.786	1.08	0.04	2554.513 ± 308.381	3.24	0.01
GS	4569.230 ± 404.874	6.25	0.01	619.041 ± 175.505	1.12	0.05	4403.212 ± 467.634	6.16	0.01
GC	4339.460 ± 562.592	6.12	0.01	641.439 ± 173.300	1.14	0.05	4479.631 ± 457.538	6.17	0.01
GB	4429.723 ± 520.624	6.15	0.01	600.573 ± 177.441	1.12	0.04	4261.805 ± 534.881	6.18	0.01
GO	4376.504 ± 467.025	6.18	0.01	$\mathbf{532.534} \pm 160.935$	1.09	0.06	4442.198 ± 456.280	6.29	0.01
GSD	4387.390 ± 422.577	6.45	0.01	1306.522 ± 256.909	1.48	0.02	4407.204 ± 638.630	6.45	0.01
GSD-D	4314.578 ± 466.079	6.42	0.01	1537.768 ± 254.603	1.45	0.01	4344.428 ± 484.607	6.39	0.01
Best	*S 2461.908 ±*			*GO 532.534 ±*			*S 2542.188 ±*		

Table 5: 7×7 large. S beats SH under both HEFT (-2%) and HEFT-1 (-6%): 2-hop high-BW paths outperform 1-hop diagonals because bottleneck bandwidth dominates over relay interference. Under HEFT-1, GO wins at ≈ 542 s: overlap-based ordering best handles dense parallel transfers in the 60-task pipeline. W is consistently worst with the highest peak link utilisation, confirming that widest-path routing maximises congestion under csma_bianchi interference.

3 Cross-Experiment Comparison

3.1 Overall Best per Experiment

Experiment	HEFT (calib.)		HEFT-1		HEFT-2	
	Routing	ms (s)	Routing	ms (s)	Routing	ms (s)
4×4 S	GSD-D	91.134 ± 0.000	GSD-D	18.035 ± 0.000	GSD-D	91.134 ± 0.000
4×4 L	GSD	855.880 ± 0.000	GS	292.091 ± 0.000	GSD	855.880 ± 0.000
7×7 S	S	93.654 ± 0.469	GSD-D	19.779 ± 0.000	SH	93.819 ± 0.577
7×7 L	S	2461.908 ± 271.335	GO	532.534 ± 160.935	S	2542.188 ± 256.661

Table 6: Best routing scheme per scheduler and experiment (mean ± std over 30 seeds). HEFT-1 achieves the lowest makespans in every experiment.

3.2 HEFT-1: Routing Rankings

Routing	4×4 S	4×4 L	7×7 S	7×7 L	**Wins**
W	18.336	497.788	24.172	1346.329	0
S	18.336	292.425	20.262	618.534	0
SH	18.336	292.425	20.262	665.738	0
GS	18.336	**292.091**†	20.262	619.041	1
GC	18.336	292.091	20.262	641.439	0
GB	18.336	292.091	20.262	600.573	0
GO	18.336	292.091	20.262	**532.534**†	1
GSD	18.336	671.422	20.262	1306.522	0
GSD-D	**18.035**	322.536	**19.779**†	1537.768	2

Table 7: HEFT-1 routing rankings across experiments. **Bold green**: global winner. **Bold**†: column winner.

3.3 HEFT (calibrated): Routing Rankings

Routing	4×4 S	4×4 L	7×7 S	7×7 L	**Wins**
W	1814.518	2383.795	1926.814	5638.899	0
S	93.856	1071.723	**93.654**†	**2461.908**†	2
SH	93.856	1071.723	93.984	2537.143	0
GS	93.319	1026.909	1301.270	4569.230	0
GC	93.319	1066.630	1151.118	4339.460	0
GB	93.319	1156.735	1503.555	4429.723	0
GO	93.319	1022.295	1506.802	4376.504	0
GSD	140.483	**855.880**†	1575.082	4387.390	1
GSD-D	**91.134**	1037.424	1381.950	4314.578	1

Table 8: HEFT (calib.) routing rankings across experiments. **Bold green**: global winner. **Bold**†: column winner.

3.4 HEFT-2: Routing Rankings

Routing	4×4 S	4×4 L	7×7 S	7×7 L	Wins
W	1814.518	2383.795	1926.814	5797.983	0
S	93.856	1071.723	93.902	**2542.188**†	1
SH	93.856	1071.723	**93.819**†	2554.513	1
GS	93.319	1026.909	1408.704	4403.212	0
GC	93.319	1066.630	1305.283	4479.631	0
GB	93.319	1156.735	1495.158	4261.805	0
GO	93.319	1022.295	1441.443	4442.198	0
GSD	140.483	**855.880**†	1327.283	4407.204	1
GSD-D	**91.134**	1037.424	1280.647	4344.428	1

Table 9: HEFT-2 routing rankings across experiments. **Bold green**: global winner. **Bold**†: column winner.

3.5 SH vs S: Does Hop-Count Routing Help?

Experiment	S (s)	SH (s)	SH vs S (%)
HEFT (calib.)			
4×4 S	93.856	**93.856**	0.0
4×4 L	1071.723	**1071.723**	0.0
7×7 S	**93.654**	93.984	+0.4
7×7 L	**2461.908**	2537.143	+3.1
HEFT-1			
4×4 S	18.336	**18.336**	0.0
4×4 L	292.425	**292.425**	0.0
7×7 S	20.262	**20.262**	0.0
7×7 L	**618.534**	665.738	+7.6
HEFT-2			
4×4 S	93.856	**93.856**	0.0
4×4 L	1071.723	**1071.723**	0.0
7×7 S	93.902	**93.819**	−0.1
7×7 L	**2542.188**	2554.513	+0.5

Table 10: SH vs S makespan comparison. **Bold**: lower of the two. SH does not win by a meaningful margin in any case where the two differ.

4 Analysis

4.1 Why HEFT-1 Dominates on csma_bianchi Grids

HEFT-1's 0.001 MB/s penalty forces all communicating tasks onto the same node or direct neighbours. On a csma_bianchi grid, PHY rates drop steeply with distance and relay hops accumulate interference; co-location avoids both. HEFT and HEFT-2 spread tasks across the topology, where runtime interference is typically 5–10× worse than HEFT's estimates. The extra-metric tables confirm this: HEFT and HEFT-2 show consistently higher peak link utilisation and mean hops than HEFT-1, which achieves near-zero utilisation by routing most transfers as same-node handoffs.

4.2 Why SH Does Not Beat S on These Grids

The hypothesis was: fewer relay hops $\Rightarrow$ fewer interfering transmitters $\Rightarrow$ lower total interference $\Rightarrow$ lower makespan. The empirical result contradicts this on the 40 m grids. On a grid with 8.6 MB/s cardinal links and lower-BW diagonal links at 56.6 m, the path that minimises $\sum 1/b$ (S) typically uses 2-hop cardinal routes rather than 1-hop diagonal routes. The 2-hop path has higher bottleneck bandwidth (8.6 vs lower diagonal BW), which more than compensates for the second active link.

When SH might win: topologies where all alternate paths have the same bandwidth (equal-BW grids) but different hop counts, making the relay interference cost the binding constraint. On the heterogeneous-BW grids tested here, this condition is not met.

4.3 Original HEFT vs HEFT-2: When Does Calibration Matter?

On the 4×4 grid, original HEFT and HEFT-2 produce identical results for all routing schemes. The topology is compact enough that different routing models return similar path bandwidths.

On the 7×7 grid with the large DAG, calibration starts to matter: HEFT (calibrated to GSD) gives $\approx$ 4192 s for GSD-routing, while HEFT-2 gives $\approx$ 4460 s. The calibrated scheduler places tasks at nodes where GSD-style congestion avoidance is effective; HEFT-2's widest-path calibration places them differently, and GSD cannot recover.

4.4 GO Wins on 7×7 Large (HEFT-1)

Under HEFT-1, tasks are placed on direct neighbours. The 60-task pipeline creates many simultaneous transfers at each stage transition. GO (overlap-based ordering) routes the most contested flows first—flows whose time windows overlap with the most other flows get first pick of low-interference paths. On the denser 7×7 grid (240 links, more routing diversity), this ordering extracts more benefit than start-time ordering (GS), delivering $\approx$ 542 s vs $\approx$ 661 s (-18%).

4.5 Key Takeaways

1. **HEFT-1 dominates on csma_bianchi grids.** Co-location is optimal; HEFT and HEFT-2 spread tasks into heavy interference, yielding 3–10$\times$ higher makespans on large experiments.

2. **Use GO with HEFT-1 on large grids.** Overlap-based ordering outperforms start-time (GS) and byte-size (GB) by routing the most contested flows first. On small grids or small DAGs, GSD-D is slightly better via deferral.

3. **SH does not improve over S** on heterogeneous-BW grids. Min-delay (S) beats min-hop (SH) because 2-hop high-BW paths deliver higher bottleneck bandwidth than 1-hop low-BW diagonals.

4. **W is always the worst routing scheme.** Widest-path routes flows through the highest-BW but also most-congested paths, maximising interference; peak link utilisation under W is consistently the highest in every experiment.

5. **Original HEFT calibration matters on large topologies.** On 7×7, calibrated HEFT outperforms HEFT-2 for dynamic routing schemes (GSD) by correctly anticipating per-scheme path bandwidths.

6. **Std devs are low under HEFT-1**, reflecting the stability of co-location: routing is irrelevant when all tasks share a node. Under HEFT/HEFT-2, std devs are larger, indicating sensitivity to per-seed topology variation (random node capacities and data sizes).

5 Reproducing These Results

```
cd ncsim/
python run_routing_eval.py
# Output: /tmp/ncsim_full_eval/  (per-seed run directories)
# Runtime: ~590 s  (3240 runs, 8 parallel workers)

python compute_interference_metrics.py
# Output: /tmp/ncsim_full_eval/grid_augmented.json

python gen_eval_results_report.py
# Output: docs/eval_results.tex
```